



Electrical Data Sheet for Socket SP5 and SP6 Processors

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	ESPI1_DAT[0]/SPI1_DAT[0]/AGPIO131, ESPI1_DAT[1]/SPI1_DAT[1]/AGPIO132, ESPI1_DAT[2]/SPI1_DAT[2]/AGPIO133, ESPI1_DAT[3]/SPI1_DAT[3]/AGPIO134, ESPI_CLK0/SPI_CLK0/AGPIO117, ESPI_CLK1/SPI_CLK1/AGPIO75, ESPI_CLK2/AGPIO74, ESPI_CS0_L/AGPIO124, ESPI_CS1_L/AGPIO125, ESPI_RSTIN_L/KBRST_L/AGPIO129, AGPIO104, AGPIO105, AGPIO106, AGPIO107, GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89, SYS_RESET_L/AGPIO1, WAKE_L/AGPIO2, THERMTRIP_L, GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89, ESPI_RSTIN_L/KBRST_L/AGPIO129, SGPIO_DATAOUT/AGPIO260, SGPIO_LOAD/AGPIO261, AGPIO256/SGPIO0_CLK, AGPIO257/SGPIO1_CLK/CLK_REQ01_L, AGPIO258/SGPIO2_CLK/CLK_REQ02_L, AGPIO259/SGPIO3_CLK, USB00_OC_L/AGPIO264, USB01_OC_L/AGPIO265, USB10_OC_L/AGPIO16, USB11_OC_L/AGPIO17, PWROK/SVI_RST_L, BP[3:0], XTRIG_L[7:4], SA[1:0], AZ_BITCLK, AZ_RST_L/SW_MDATA1, AZ_SDIN0/SW_MDATA3, AZ_SDIN1/SW_MCLK, AZ_SDIN2/SW_MDATA0, AZ_SDOOUT/SW_MDATA2, AZ_SYNC, AGPIO109, AGPIO115/CLK_REQ11_L, AGPIO116/CLK_REQ12_L, AGPIO21, AGPIO22, AGPIO3/SPD_HOST_CTRL_L, AGPIO4/SATA_ACT_L, AGPIO5/DEVSLP0/SATA_ZP0_L, AGPIO6/DEVSLP1/SATA_ZP1_L, AGPIO76/SPI_TPM_CS_L, AGPIO88, DBREQ_L, PCIE_RST0_L/AGPIO266, PCIE_RST1_L/AGPIO26, ESPI0_ALERT_L/AGPIO108, ESPI_RSTOUT_L/AGPIO23, NMI_SYNC_FLOOD_L, PWRGD_OUT/AGPIO12, and RTCCLK	23
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Revision History

Date	Revision	Description
January 2025	0.83	Updated Chapter 3.
January 2024	0.82	Updated Figure 3 and Table 16.
October 2023	0.81	Added Section 1.1. Updated Table 16. Updated Table 17 notes. Added note in Chapter 16.
August 2023	0.80	Revised Table 1.
July 2023	0.78	Updated Table 1.
March 2023	0.77	Added I3C pin parameters in Table 10.
February 2023	0.76	Added support for Family 1Ah Models 00h–0Fh processors. Added Bidirectional pin parameters. Updated eSPI timing information. Updated Figure 11.
September 2022	0.75	Added support for Family 19h Models A0h–AFh processors. Updated JTAG timing, eSPI timing, reset sequencing, and power sequencing information.
April 2022	0.74	Added Table 7 and Table 11. Updated Chapter 3.
February 2022	0.73	Corrected t_2 description in Table 21.
December 2021	0.72	Added support for Family 19h Models 10h–1Fh Socket SP6 processors.
November 2021	0.71	Updated eSPI output timing. Added GPIO maximum drive strength specification. Updated Bidirectional pin list.
August 2021	0.70	Updated I3C and Bidirectional pin lists. Added note regarding PCIe reference clock buffer.
April 2021	0.60	Updated I3C tables, reset sequencing, and power sequencing information.
February 2021	0.50	Initial NDA version.

Chapter 1 Introduction

This document contains power sequencing requirements and various I/O electrical requirements for the SP5 and SP6 infrastructures.

- Socket SP5
 - Family 19h Models 10h–1Fh
 - Family 19h Models A0h–AFh
 - Family 1Ah Models 00h–0Fh
 - Family 1Ah Models 10h–1Fh
- Socket SP6
 - Family 19h Models A0h–AFh

1.1 Reference Documents

For additional information, refer to the following AMD publications to supplement this document:

- *Infrastructure Roadmap (IRM)*
 - *Infrastructure Roadmap (IRM) for Socket SP5 Processors*, publication #56943
 - *Infrastructure Roadmap (IRM) for Socket SP6 Processors*, publication #57240
- *Functional Data Sheet (FDS)*
 - *Socket SP5 Processor Functional Data Sheet*, publication #56900
 - *Socket SP6 Processor Functional Data Sheet*, publication #57388
- *Motherboard Design Guide (MBDG)*
 - *Socket SP5 Processor Motherboard Design Guide*, publication #56870
 - *Socket SP6 Processor Motherboard Design Guide*, publication #57387
- *Processor Programming Reference (PPR)*
 - *Processor Programming Reference (PPR) for AMD Family 19h Models 10h–1Fh Processors*, publication #55901
 - *Processor Programming Reference (PPR) for AMD Family 19h Models A0h–AFh Processors*, publication #57228
 - *Processor Programming Reference (PPR) for AMD Family 1Ah Models 00h–0Fh Processors*, publication #57238
 - *Processor Programming Reference (PPR) for AMD Family 1Ah Models 10h–1Fh Processors*, publication #57883
- *Memory Population Guidelines*

- *Memory Population Guidelines for AMD Family 19h Models 10h–1Fh and Models A0h–AFh Socket SP5 Processors*, publication #56960
- *Memory Population Guidelines for AMD Family 19h Models A0h–AFh Socket SP6 Processors*, publication #57609
- *Memory Population Guidelines for AMD Family 1Ah Models 00h–0Fh and Models 10h–1Fh Socket SP5 Processors*, publication #58266
- *AMD Serial VID Interface Version 3 (SVI3) Specification*, publication #56413

Chapter 2 Absolute Maximum Ratings

Table 1 specifies absolute maximum ratings. Absolute maximum ratings lie outside the operating limits of the processor. Long-term reliability and functionality are expected within operating conditions. Systems using this device must be designed to ensure that absolute maximum ratings are not violated.

At conditions outside the operating limits of the processor, but within the absolute maximum ratings, functionality and long-term reliability of the device are not guaranteed or expected. If the device is returned to within the operating conditions, the device may function, but the long-term reliability may be degraded.

Device functionality and reliability will be compromised when a device is operated outside the absolute maximum ratings. Furthermore, when a device is returned to within operating conditions the device may not work and its long-term reliability will have been degraded.

Table 1. Absolute Maximum Ratings

Characteristic	Range
VDDCR_CPU0 and VDDCR_CPU1 supply voltages relative to VSS	–0.3 to 1.70 V
VDDCR_SOC supply voltage relative to VSS	–0.3 to 1.35 V
VDDIO supply voltage relative to VSS	–0.3 to 1.28 V
VDDIO_AUDIO supply voltage relative to VSS	–0.3 to 1.98 V
VDDBT_RTC_G supply voltage relative to VSS	–0.3 to 3.30 V
VDD_11_S3 supply voltage relative to VSS	–0.3 to 1.20 V
VDD_18_S5 supply voltage relative to VSS	–0.3 to 1.98 V
VDD_33_S5 supply voltage relative to VSS	–0.3 to 3.63 V

Chapter 3 DDR SDRAM Interface

Silicon is DDR5 JEDEC compliant.

Socket SP5 supports transfer rates from 3600 to 6400 MT/s, and Socket SP6 supports transfer rates from 3600 to 4800 MT/s. For more information, refer to the *Memory Population Guidelines* documents.

Chapter 4 PCI Express® Interface

Refer to the *PCI Express Base Specification Revision 5.0, Version 1.0*. Any deviation from the specifications is listed in this document.

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Chapter 5 xGMI Interface

The xGMI receivers support PCIe Gen5 compatible channels.

Contact your local AMD representative for simulation models.

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Chapter 6 Clock Pins

For additional information on common mode, refer to the *PCI Express Base Specification*.

Table 2. Combined AC and DC Operating Conditions for GPP_CLK01N, GPP_CLK01P, GPP_CLK02N, GPP_CLK02P, GPP_CLK03N, GPP_CLK03P, GPP_CLK04N, GPP_CLK04P, GPP_CLK05N, GPP_CLK05P, REFCLK100SSC_N/GPP_CLK10N, REFCLK100SSC_P/GPP_CLK10P, GPP_CLK11N, GPP_CLK11P, GPP_CLK12N, GPP_CLK12P, GPP_CLK13N, GPP_CLK13P, GPP_CLK14N, GPP_CLK14P, GPP_CLK15N, and GPP_CLK15P

Symbol	Parameter	Min	Max	Unit
freq	Frequency (output)	48	100	MHz
t _{Fall/Rise}	Measured from -150 mV to +150 mV	0.6	4.0	V/ns
V _{IH}	Differential input high voltage	150	—	mV
V _{IL}	Differential input low voltage	—	-150	
V _{Cross}	Absolute crossing point voltage	250	550	
V _{Crossdelta}	Variation of V _{Cross} over all rising clock edges	—	140	
—	Duty cycle	40	60	%
—	Output driver termination	40	60	Ω

Notes:

1. External input clock is supported on GPP_CLK02N, GPP_CLK02P, GPP_CLK12N, GPP_CLK12P.
2. When a buffer is added to fan out the PCIe reference clock, choose buffers that contribute a minimum amount of jitter. AMD recommends a fan out buffer with a jitter of less than 20 fSec.

Table 3. Combined AC and DC Operating Conditions for X48M_X1 and X48M_X2

Symbol	Parameter	Min	Typical	Max	Unit	Notes
$f_{\text{xoclk_48}}$	Crystal or external clock generator reference frequency	—	48	—	MHz	3, 4
f_{acc}	External clock generator output frequency accuracy	-50	—	50	PPM	1, 4, 7
C_{In}	Input capacitance	8.4	—	8.4	pF	
D_{L}	Drive level	—	200	600	μW	4
ESR	Equivalent series resistance	—	10	50	Ω	4
V_{IH}	Input high level	1.62	1.8	—	V	5
V_{IL}	Input low level	—	0	0.18		5
DC	Duty cycle	45	50	55	%	5
t_{IR}	48 MHz rise slew rate	1	—	—	V/ns	5, 6
t_{IF}	48 MHz fall slew rate	1	—	—		5, 6
t_{Short}	Short term peak-to-peak jitter	—	—	200	ps	5
t_{IJLT}	48 MHz long term jitter requirement (10 μs after scope trigger)	—	—	250		5
$V_{\text{i_low_PAD_XTALIN}}$	Voltage level of logic LOW of crystal/oscillator signal PAD_XTALIN. Referenced to VSS_XTAL	—	0	—	V	4
$V_{\text{i_high_PAD_XTAQLI}}$	Voltage level of logic HIGH of crystal/oscillator	1.62	—	1.98		4

Notes:

1. $25^{\circ}\text{C} \pm 3^{\circ}\text{C}$
2. $0\text{--}85^{\circ}\text{C}$
3. Maximum 30 PPM
4. Crystal mode
5. External oscillator mode
6. For acceptable duty-cycle performance, t_{IR} and t_{IF} should be matched to within 10% of each other
7. This is the maximum PPM tolerated by the processor; but other factors such as time-keeping accuracy within the overall system may require a smaller PPM. Refer to the “Socket SP5 Processor Motherboard Design Guide”, publication #56870, for suggested PPM values.

Table 4. Combined AC and DC Operating Conditions for X32K_X1 and X32K_X2

Symbol	Parameter	Min	Typical	Max	Unit	Notes
$f_{\text{xoclk_32}}$	Crystal reference frequency	—	32.768	—	KHz	
f_{acc}	External clock generator output frequency accuracy	-100	—	100	PPM	1, 2
C_{In}	Input capacitance	9.0	—	9.5	pF	5
V_{IH}	Input high	$0.8 \times V_{\text{DDBT_RTC_G}}$	—	—	V	3, 4
V_{IL}	Input low	—	—	$0.2 \times V_{\text{DDBT_RTC_G}}$		3, 4
t_{IR}	32 kHz rise slew rate	0.066	—	—	V/ns	3, 4
t_{IF}	32 kHz fall slew rate	0.066	—	—		3, 4
t_{IJLT}	32 kHz long term jitter requirement	—	—	100	ns	3, 4
t_{Short}	Short term peak-to-peak jitter	—	—	100		3, 4
—	Duty cycle	48	50	52	%	3, 4

Notes:

1. Crystal mode
2. This is the maximum PPM tolerated by the processor, but other factors such as time-keeping accuracy within the overall system may require a smaller PPM. Refer to the “Socket SP5 Processor Motherboard Design Guide”, publication #56870, for suggested PPM values.
3. $V_{\text{DDBT_RTC_G}} = 1.5$ or 1.8 V
4. In external oscillator mode:
If the CLK from RTC IC is designed as G3 domain, you can connect it to X32K_X2 to save battery power.
If the CLK from RTC IC is designed as G3 domain, you can connect it to X32K_X1 to avoid battery power leakage to the pull high resistor.
5. In Crystal mode, C_{In} could be adjusted to get least PPM error.

Table 5. Combined AC and DC Operating Conditions for REFCLK100SSC_N and REFCLK100SSC_P

Symbol	Parameter	Min	Max	Unit
freq	Frequency (input and output)	48	100	MHz
$t_{\text{Fall/Rise}}$	Measured from -150 mV to +150 mV	0.6	4.0	V/ns
V_{IH}	Differential input high voltage	150	—	mV
V_{IL}	Differential input low voltage	—	-150	
V_{Cross}	Absolute crossing point voltage	250	550	
$V_{\text{Crossdelta}}$	Variation of V_{Cross} over all rising clock edges	—	140	
—	Duty cycle	40	60	%
—	Output termination	40	60	Ω

Chapter 7 Miscellaneous Pins

For input and bidirectional pins:

- Maximum V_{IH-DC} is the VDD voltage.
- Maximum V_{IH-AC} is 110% of the VDD voltage; overshoot is limited in duration to 25% of the clock period of the highest frequency of the pin's clock.
- Minimum V_{IL-DC} is ground.
- Minimum V_{IL-AC} is 10% of the VDD voltage below ground; undershoot is limited in duration to 25% of the clock period of the highest frequency of the pin's clock.
- For GPIO, VDD_18, VDD_18_S5 = 1.8 V.
- GPIO maximum drive strength is 8.34 mA.

7.1 Miscellaneous Input Pins

Table 6. Combined AC and DC Operating Conditions for RSMRST_L, TCK, TDI, TMS, TRST_L, DBREQ_L, FORCE_SELFREFRESH, PWR_GOOD, PROCHOT_L, PWR_BTN_L/AGPIO0, PCC0_L, and PCC1_L

Symbol	Parameter	Min	Max	Unit	Notes
V _{In}	Max input voltage	—	1.80	V	
V _{IH-DC 1.8V RX}	Input high voltage (logic 1)	1.20	1.80		
V _{IH-AC 1.8V RX}			1.98		1
V _{IL-DC 1.8V RX}	Input low voltage (logic 0)	0	0.70		
V _{IL-AC 1.8V RX}		-0.18			1
I _L	Input leakage current	—	5	μA	
C _{In}	Input capacitance	—	1.5	pF	
t _{PROCHOT_L}	PROCHOT_L assertion time	5	—	ms	

Notes:

1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

Table 7. Combined AC and DC Operating Conditions for RTC_LDO_SELECT

Symbol	Parameter	Min	Max	Unit
V _{In}	Max input voltage	—	3.3	V
V _{IH-DC}	Input high voltage (logic 1)	2.2	3.3	
V _{IH-AC}			3.46	
V _{IL-DC}	Input low voltage (logic 0)	0	0.7	
V _{IL-AC}		-0.3		
I _L	Input leakage current	—	5	uA
C _{In}	Input capacitance	—	1.5	pF

For TCK, TDI, and TMS timing, see Chapter 17.

7.2 Miscellaneous Output Pins

Table 8. Combined AC and DC Operating Conditions for TDO, SLP_S3_L, SLP_S5_L, AGPIO7, SMERR_L/AGPIO24, and NV_SAVE_L

Symbol	Parameter	Min	Max	Unit
V_{OH-DC}	DC output high voltage (logic 1) @ $I = 2$ mA	$VDD_{18_S5} - 0.45$	—	V
V_{OL-DC}	DC output low voltage (logic 0) @ $I = -2$ mA	—	0.45	

For TDO timing, see Chapter 17.

7.3 Miscellaneous Bidirectional Pins

**Table 9. Combined AC and DC Operating Conditions for I²C/AGPIO Pins:
I2C4_SCL/HP_SCL/UBM_SCL/AGPIO13,
I2C4_SDA/HP_SDA/UBM_SDA/AGPIO14,
I2C5_SCL/BMC_SCL/SMBUS1_SCL/AGPIO19, and
I2C5_SDA/BMC_SDA/SMBUS1_SDA/AGPIO20**

Symbol	Parameter	Min	Max	Unit	Notes
V _{OH-DC} 1.8V Mode	DC output high voltage (logic 1) @ I = 2 mA	—	—	V	
V _{OL-DC} 1.8V Mode	DC output low voltage (logic 0) @ I = −2 mA	—	0.36		
V _{IH-DC} 1.8V Mode	Input high voltage (logic 1)	1.26	1.80		
V _{IH-AC} 1.8V Mode			1.98		1
V _{IL-DC} 1.8V Mode	Input low voltage (logic 0)	0	0.54		
V _{IL-AC} 1.8V Mode		-0.18			1
I _L 1.8V Mode	Input leakage current	—	10	μA	
C _{In} 1.8V Mode	Input capacitance	—	10	pF	

Notes:

1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

The I²C interface can function in 1.8 V mode. For further details, refer to the *Functional Data Sheet (FDS)* and *Motherboard Design Guide (MBDG)* documents.

Table 10. Combined AC and DC Operating Conditions for I³C Pins:

**I3C0_SCL/I2C0_SCL/SPD0_SCL/SMBUS0_SCL/AGPIO145,
I3C0_SDA/I2C0_SDA/SPD0_SDA/SMBUS0_SDA/AGPIO146,
I3C1_SCL/I2C1_SCL/SPD1_SCL/AGPIO147, I3C1_SDA/I2C1_SDA/SPD1_SDA/AGPIO148,
I3C2_SCL/I2C2_SCL/SPD2_SCL/AGPIO149, I3C2_SDA/I2C2_SDA/SPD2_SDA/AGPIO150,
I3C3_SCL/I2C3_SCL/SPD3_SCL/AGPIO151, and
I3C3_SDA/I2C3_SDA/SPD3_SDA/AGPIO152**

Symbol	Parameter	Min	Max	Unit	Notes
V_{OH} 1.1V Mode	Output high voltage (logic 1) @ $I = 4$ mA	$0.75 \times VDD_11_S3$	—	V	
V_{OL} 1.1V Mode	Output low voltage (logic 0) @ $I = 4$ mA	—	$0.25 \times VDD_11_S3$		
V_{IH-DC} 1.1V Mode	Minimum DC voltage at PAD pin that will produce a logic high output on 1.1V receiver	0.76	—		
V_{IH-AC} 1.1V Mode		0.92	—		
V_{IL-DC} 1.1V Mode	Maximum DC voltage at PAD pin that will produce logic low output on 1.1V receiver	—	0.32		
V_{IL-AC} 1.1V Mode		—	0.095		
I_L 1.1V Mode	Input leakage current (external pull-up of 3 k Ω) $VDD = VDD_11_S3$	—	5	μA	
C_{In} 1.1V Mode	Input capacitance $VDD = VDD_11_S3$	—	5	pF	
V_{HYS}	Input hysteresis	100	—	mV	
V_{IH_Min}	Input high voltage	$0.7 \times VDD_11_S3$	—	V	
V_{IL_Max}	Input low voltage	—	$0.3 \times VDD_11_S3$		
V_{SLPRV}	Receive buffer slope reversal	150	—	mV _{PK-PK}	1

Notes:

1. Slope reversal that occurs between V_{IH_Min} and V_{IL_Max} on either the rising or the falling edge, must be less than V_{SLPRV} (see Figure 1a). Rising edge slope reversal, where the signal first rises above V_{IH_Min} and then falls below V_{IH_Min} , must not drop more than V_{SLPRV} below V_{IH_Min} (see Figure 1b). Falling edge slope reversal, where the signal first drops below V_{IL_Max} then rises above V_{IL_Max} , must not rise more than V_{SLPRV} above V_{IL_Min} (see Figure 1c).

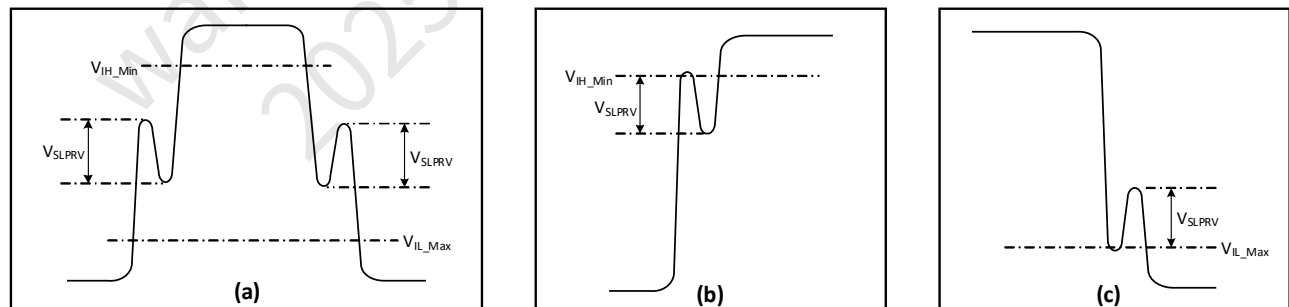
**Figure 1. Slope Reversal Specification**

Table 11. Combined AC and DC Operating Conditions for SEC_SCL and SEC_SDA

Symbol	Parameter	Min	Max	Unit	Notes
V _{IH-DC} 1.1V Mode	Minimum DC voltage at PAD pin that will produce a logic high output on 1.1 V receiver	0.76	—	V	
V _{IH-AC} 1.1V Mode		0.92	—		
V _{IL-DC} 1.1V Mode	Maximum DC voltage at PAD pin that will produce logic low output on 1.1 V receiver	—	0.32		
V _{IL-AC} 1.1V Mode		—	0.095		
V _{OH} 1.1V Mode	Output high voltage (logic 1) @ I = 4 mA	0.75×VDD_11_S3	—		
V _{OL} 1.1V Mode	Output low voltage (logic 0) @ I = 4 mA	—	0.25×VDD_11_S3		
V _{IH-DC} 1.8V Mode	Input high voltage (logic 1)	1.26	1.80		
V _{IH-AC} 1.8V Mode			1.98		1
V _{IL-DC} 1.8V Mode	Input low voltage (logic 0)	0	0.54		
V _{IL-AC} 1.8V Mode		-0.18			1
V _{OH-DC} 1.8V Mode	DC output high voltage (logic 1) @ I = 2 mA	—	—		
V _{OL-DC} 1.8V Mode	DC output low voltage (logic 0) @ I = −2 mA	—	0.36		
I _L 1.1V Mode	Input leakage current (external pull-up of 3 kΩ) VDD = VDD_11_S3	—	5	μA	
I _L 1.8V Mode	Input leakage current	—	10		
C _{In} 1.1V Mode	Input capacitance VDD = VDD_11_S3	—	5	pF	
C _{In} 1.8V Mode	Input capacitance	—	10		
V _{HYS}	Input hysteresis	100	—	mV	

Notes:

1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

Table 12. Combined AC and DC Operating Conditions for Bidirectional Pins:
 UART0_CTS_L/UART2_TXD/AGPIO135, UART0_INTR/AGPIO139,
 UART0_RTS_L/UART2_RXD/AGPIO137, UART0_RXD/AGPIO136,
 UART0_TXD/AGPIO138, UART1_RXD/AGPIO141, UART1_TXD/AGPIO142,
 RESET_L, SPI_CS0_L/AGPIO118, SPI_CS1_L/AGPIO119, SPI_CS2_L/AGPIO126,
 ESPI0_DAT[0]/SPI0_DAT[0]/AGPIO120, ESPI0_DAT[1]/SPI0_DAT[1]/AGPIO121,
 ESPI0_DAT[2]/SPI0_DAT[2]/AGPIO122, ESPI0_DAT[3]/SPI0_DAT[3]/AGPIO123,
 ESPI1_DAT[0]/SPI1_DAT[0]/AGPIO131, ESPI1_DAT[1]/SPI1_DAT[1]/AGPIO132,
 ESPI1_DAT[2]/SPI1_DAT[2]/AGPIO133, ESPI1_DAT[3]/SPI1_DAT[3]/AGPIO134,
 ESPI_CLK0/SPI_CLK0/AGPIO117, ESPI_CLK1/SPI_CLK1/AGPIO75,
 ESPI_CLK2/AGPIO74, ESPI_CS0_L/AGPIO124, ESPI_CS1_L/AGPIO125,
 ESPI_RSTIN_L/KBRST_L/AGPIO129, AGPIO104, AGPIO105, AGPIO106, AGPIO107,
 GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89,
 SYS_RESET_L/AGPIO1, WAKE_L/AGPIO2, THERMTRIP_L,
 GENINT_L/PM_INTR_L/UBM_CPRSNT_CHANGE_DET_L/AGPIO89,
 ESPI_RSTIN_L/KBRST_L/AGPIO129, SGPIO_DATAOUT/AGPIO260,
 SGPIO_LOAD/AGPIO261, AGPIO256/SGPIO0_CLK,
 AGPIO257/SGPIO1_CLK/CLK_REQ01_L, AGPIO258/SGPIO2_CLK/CLK_REQ02_L,
 AGPIO259/SGPIO3_CLK, USB00_OC_L/AGPIO264, USB01_OC_L/AGPIO265,
 USB10_OC_L/AGPIO16, USB11_OC_L/AGPIO17, PWROK/SVI_RST_L, BP[3:0],
 XTRIG_L[7:4], SA[1:0], AZ_BITCLK, AZ_RST_L/SW_MDATA1,
 AZ_SDIN0/SW_MDATA3, AZ_SDIN1/SW_MCLK, AZ_SDIN2/SW_MDATA0,
 AZ_SDOUT/SW_MDATA2, AZ_SYNC, AGPIO109, AGPIO115/CLK_REQ11_L,
 AGPIO116/CLK_REQ12_L, AGPIO21, AGPIO22, AGPIO3/SPD_HOST_CTRL_L,
 AGPIO4/SATA_ACT_L, AGPIO5/DEVSLP0/SATA_ZP0_L,
 AGPIO6/DEVSLP1/SATA_ZP1_L, AGPIO76/SPI_TPM_CS_L, AGPIO88, DBREQ_L,
 PCIE_RST0_L/AGPIO266, PCIE_RST1_L/AGPIO26, ESPI0_ALERT_L/AGPIO108,
 ESPI_RSTOUT_L/AGPIO23, NMI_SYNC_FLOOD_L, PWRGD_OUT/AGPIO12, and
 RTCCLK

Symbol	Parameter	Min	Max	Unit	Notes
V _{OH-DC}	DC output high voltage (logic 1) @ I = 2 mA	VDD_18 – 0.45	—	V	1
V _{OL-DC}	DC output low voltage (logic 0) @ I = –2 mA	—	0.45		1
V _{IH-DC}	Input high voltage (logic 1)	0.65×VDD_18	VDD_18		2
V _{IH-AC}			1.10×VDD_18		
V _{IL-DC}	Input low voltage (logic 0)	0	0.35×VDD_18		2
V _{IL-AC}		0.10×(–VDD_18)			
V _{hys}	Input hysteresis	150	—	mV	
I _L	Input leakage current	—	5	μA	
C _{In}	Input capacitance	—	1.5	pF	
—	SYS_RESET_L assertion time	10	—	ms	
—	KBRST_L assertion time	0.2	50		

Notes:

1. This parameter does not apply for RESET_L.
2. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

Chapter 8 Thermal Protection

8.1 THERMTRIP_L Thermal Protection

THERMTRIP functionality is a mechanism to protect the processor from thermal damage. Assertion of THERMTRIP_L indicates the factory determined thermal shutdown temperature ($T_{DIE, SHUTDOWN}$) has been reached or exceeded. The specified power planes must be powered down within a maximum defined time (t_{SD_DELAY}) and the external chipset is expected to place the system in the S5 ACPI state (power off). Deassertion of PWROK resets the THERMTRIP_L assertion.

Additional information in regards to thermal protection circuitry reference designs and thermal solution guidelines can be found in the *Processor Programming Reference (PPR)* documents.

Table 13. Platform Thermal Protection Limits for the Processor

Symbol	Parameter Description	Typ	Max	Unit	Notes
$T_{CTL,SHUTDOWN}$	THERMTRIP shutdown temperature on T_{CTL} scale	115	—	°C	3
$T_{DIE,SHUTDOWN}$	THERMTRIP shutdown die temperature	—	125		1
t_{SD_DELAY}	Maximum allowed time from THERMTRIP_L detection to power plane shutdown to prevent thermal damage to the processor.	—	500	ms	2

Notes:

1. T_{DIE} refers to the physical temperature measured in degrees Celsius. T_{DIE} may not be reported from the processor.
2. Shutdown is specified within this context as ramping down of the power planes to 20% of the normal specified value.
3. A non-slew rate controlled T_{CTL} temperature is used to compare against THERMTRIP shutdown threshold.

Chapter 9 SVI Pins

For implementation details and electrical specifications of the SVC, SVD, SVTI, and SVTO pins, refer to the *AMD Serial VID Interface Version 3 (SVI3) Specification*, publication #56413.

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Chapter 10 SB-TSI Interface

For the non-electrical characteristics of the SB-TSI signals, refer to the “SB Temperature Sensor Interface (SB-TSI)” section in the *Processor Programming Reference (PPR)* documents.

The electrical characteristics of SB-TSI pins are listed in Table 14. Standard mode (100 KHz) and fast mode (400 KHz) are supported; high-speed mode (3.4 MHz) is not supported.

Table 14. Combined AC and DC Operating Conditions for ALERT_L

Symbol	Parameter	Min	Max	Unit
V _{OH-DC}	DC output high voltage (logic 1) @ I = 2 mA	VDD_18_S5 – 0.45	—	V
V _{OL-DC}	DC output low voltage (logic 0) @ I = –2 mA	—	0.45	

Notes:

1. Input overshoot and undershoot are limited in duration to 25% of the clock period of the highest frequency of the pin's clock.

Table 15. Combined AC and DC Operating Conditions for SIC and SID

Symbol	Parameter	Min	Max	Unit
V _{IH-DC 1.1V}	Minimum DC voltage at PAD pin that will produce a logic high output on 1.1 V receiver	0.76	—	V
V _{IH-AC 1.1V}		0.92	—	
V _{IL-DC 1.1V}	Maximum DC voltage at PAD pin that will produce logic low output on 1.1 V receiver	—	0.32	
V _{IL-AC 1.1V}		—	0.095	
I _{L 1.1V}	Input leakage current (external pull-up of 3 kΩ) VDD = VDD_11_S3	—	5	uA
C _{In 1.1V}	Input capacitance VDD = VDD_11_S3	—	5	pF
V _{HYS}	Input hysteresis	100	—	mV
V _{OH 1.1V}	Output high voltage (logic 1) @ I = 4 mA	0.75×VDD_11_S3	—	V
V _{OL 1.1V}	Output low voltage (logic 0) @ I = 4 mA	—	0.25×VDD_11_S3	

Chapter 11 SATA Interface

For electrical characteristics of the SATA interface signals, refer to the *Serial ATA Revision 3.2 Gold Specification*.

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Chapter 12 USB 3/USB 2 Interface

Silicon is USB 3 SuperSpeed compliant.

For electrical characteristics of the USB_SS interface signals, refer to the *USB 3.2 Gen1 Specification*. For electrical characteristics of the USB_HS interface signals, refer to the *USB 2.0 Specification*.

Chapter 13 SPI/eSPI Interface

13.1 SPI

The SPI bus interfaces support single, dual, and quad modes. The SPI interfaces support speeds up to 100 MHz.

Table 16. SPI Timing

S-Clock			16 MHz		22 MHz		33 MHz		66 MHz		100 MHz		Notes
Symbol	Parameter	Unit	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
t_{SLCH}	SPI_CSx_L or SPI_TPM_CS_L assertion setup	ns	60	—	43	—	28	—	13	—	8	—	1, 2, 3
t_{CHSH}	SPI_CSx_L or SPI_TPM_CS_L assertion hold	ns	60	—	43	—	28	—	13	—	8	—	1, 2, 3
t_{CLQV}	Output valid time	ns	—	3	—	3	—	3	—	3	—	3	1, 2
t_{DS}	Data in setup	ns	3.4	—	3.4	—	3.4	—	3.4	—	3.4	—	1, 2
t_{DH}	Data in hold	ns	1	—	1	—	1	—	1	—	1	—	1, 2
f_R	Clock frequency read instruction	MHz	16.34	17	21.78	23	32.67	34	65.34	67	99	101	
t_{CLH}	Clock high time	ns	26.1	34.5	19.58	25.88	13.0	17.3	6.5	8.6	3.9	6.1	
t_{CLL}	Clock low time	ns	26.1	34.5	19.58	25.88	13.0	17.3	6.5	8.6	3.9	6.1	
t_{CLCH}	Clock rise time	ns	0.1	—	0.1	—	0.1	—	0.1	—	0.1	—	
t_{CHCL}	Clock fall time	ns	0.1	—	0.1	—	0.1	—	0.1	—	0.1	—	

Notes:

1. Trace flight time is not included.
2. The output load is 10 pF.
3. Where x is 1 or 2.

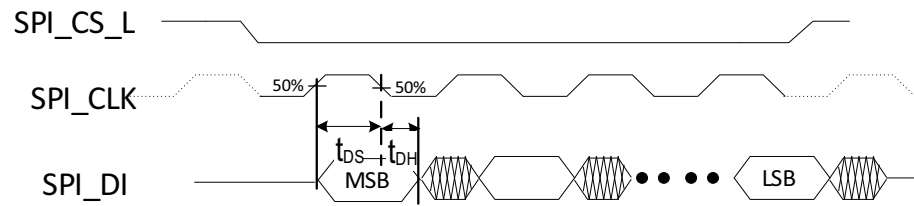


Figure 2. SPI Input Timing

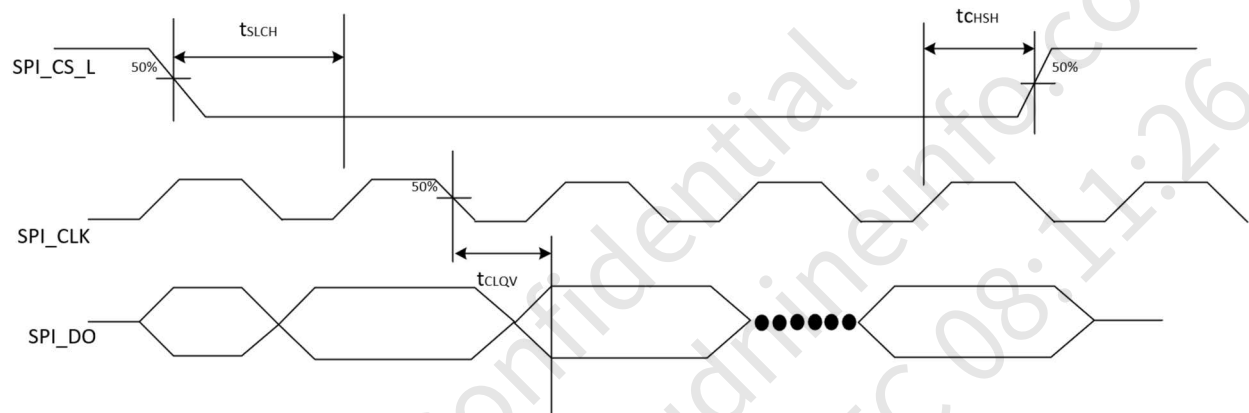


Figure 3. SPI Output Timing

13.2 eSPI

Refer to the *Interface Base Specification for the Enhanced Serial Peripheral Interface (eSPI), Revision 1.0*.

The eSPI bus interfaces support single, dual, and quad modes. The eSPI bus interfaces support speeds of 16 MHz, 20 MHz, 25 MHz, 33 MHz, 50 MHz, and 66 MHz.

Table 17. eSPI Timing

Symbol	Parameter	Unit	16 MHz		20 MHz		25 MHz		33 MHz		50 MHz		66 MHz	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
T _{CK}	Clock period	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{CKH}	Clock high time	T _{CK}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6
T _{CKL}	Clock low time	T _{CK}	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6	0.4	0.6
T _{CSST}	Chip select setup time	ns	90	—	75	—	60	—	45	—	30	—	22	—
T _{CSHT}	Chip select hold time	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{CSDT}	Chip select deassertion time	ns	60	—	50	—	40	—	30	—	20	—	15	—
T _{DIST}	Data in setup time	ns	7	—	7	—	7	—	7	—	7	—	7	—
T _{DIHT}	Data in hold time	ns	0.3	—	0.3	—	0.3	—	0.3	—	0.3	—	0.3	—
T _{ODTT}	Output disable time during turnaround	ns	—	18	—	15	—	12	—	9	—	8	—	6
T _{DOVT}	Data output valid time	ns	1	3	1	3	1	3	1	3	1	3	1	3
T _{INIT}	eSPI reset# deassertion to first transaction (GET_CONFIGURATION)	us	1	—	1	—	1	—	1	—	1	—	1	—
T _{INIT-FREQ}	Initial bus frequency upon eSPI reset# deassertion	MHz	—	20	—	20	—	20	—	20	—	20	—	20

Notes:

1. All parameters in Table 17 are specified with respect to AMD processor pins and NOT device pins as described in the eSPI Specification.
2. The output load is 10 pF.

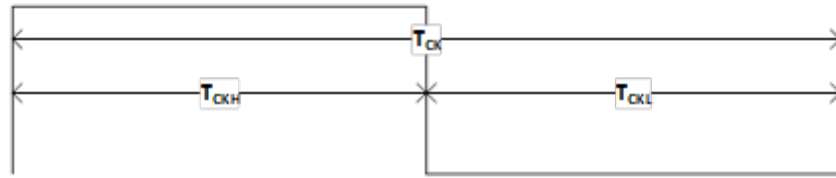


Figure 4. eSPI Clock Timing

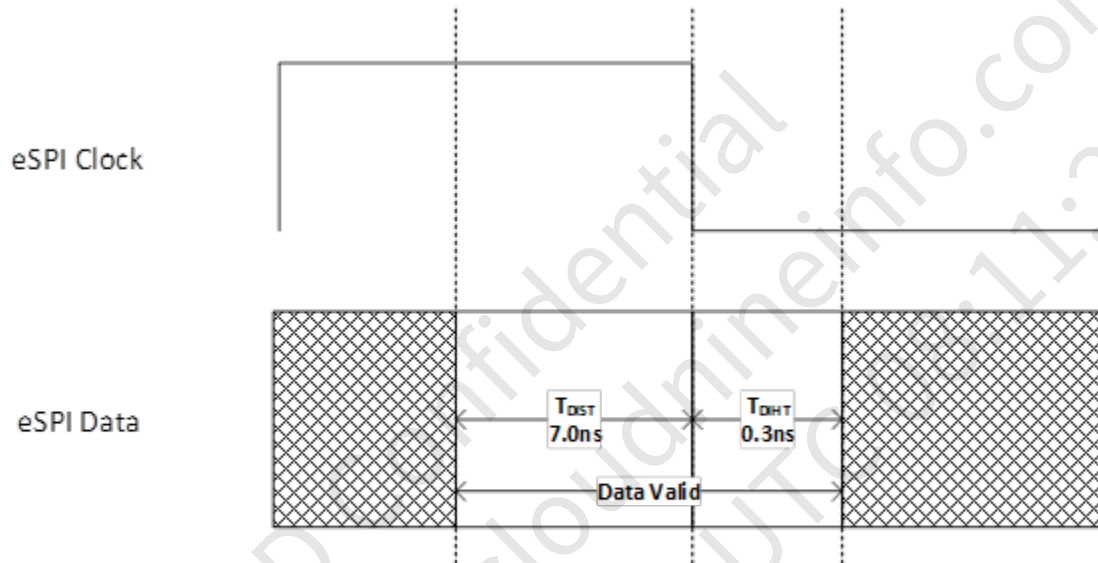


Figure 5. eSPI Input Timing

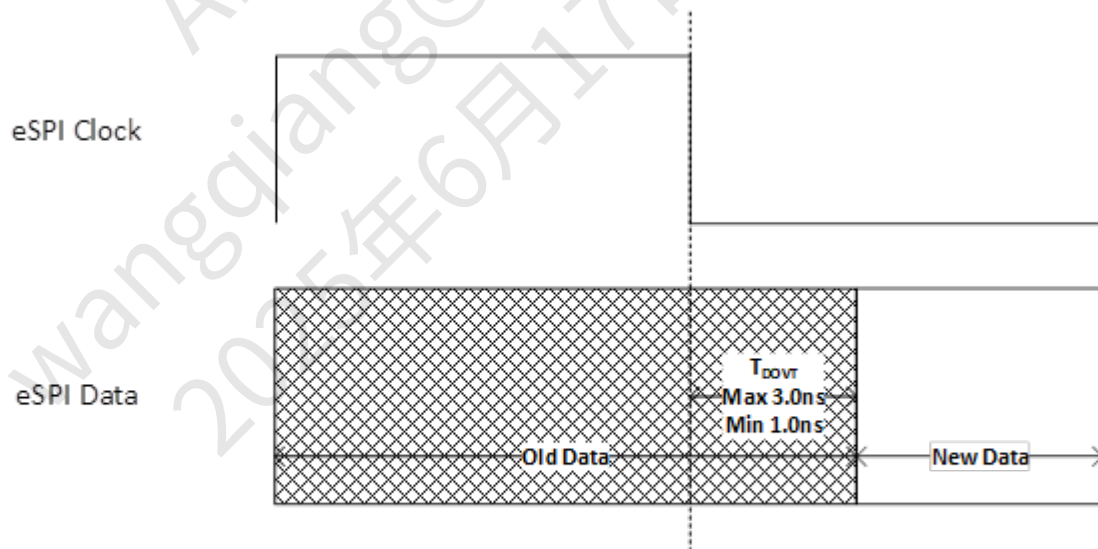
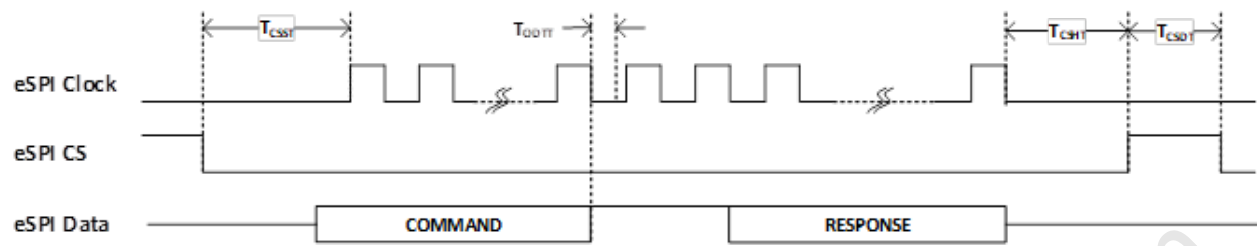


Figure 6. eSPI Output Timing

**Figure 7. eSPI Protocol Timing**

Chapter 14 System Management Bus (SMBus) Interface

For electrical characteristics of the SMBus interface signals, refer to the *System Management Bus (SMBus) Specification Version 2.0*.

Chapter 15 UART Interface

For information on register space, refer to the *Processor Programming Reference (PPR)* documents.

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Chapter 16 I²C/I³C Interface

For information on the I²C and I³C interfaces:

- Refer to the *I²C Specification, I²C Bus Specification and User Manual, UM10204 Rev. 5, 9 October 2012*.
- Refer to the *I³C Basic v1.0 Specification*¹ and Table 10 in this document.

Notes:

1. When communicating with I²C legacy devices, t_{LOW} can be relaxed to 1200 ns. For details on t_{LOW} , refer to Table 57 “I³C Timing Requirements When Communicating with I²C Legacy Devices” in the “I³C Basic v1.0 Specification”.

Chapter 17 JTAG Interface

Refer to the *IEEE Standard Test Access Port and Boundary-Scan Architecture Specification, IEEE Std 1149.1[™]-2001(R2008)*.

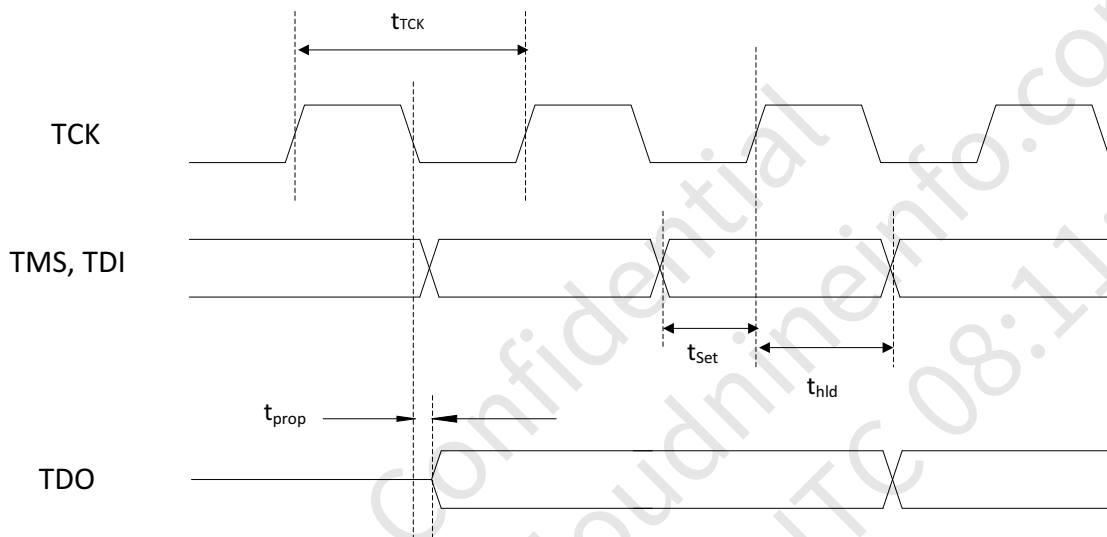


Figure 8. JTAG Timing

Table 18. JTAG Timing Parameters

Parameter	Description	Min	Max	Unit	Notes
t_{TCK}	TCK period	100	1000	ns	1
t_{prop}	TDO output propagation delay from TCK falling edge	—	40		
t_{Set}	TDI, TMS input setup to TCK rising edge	40	—		
t_{hld}	TDI, TMS input hold from TCK rising edge	40	—		

Notes:

1. t_{TCK} may be held low indefinitely.

Chapter 18 Power Supplies and Signal Sequencing

18.1 Power Supply Relationships

Power supply relationships during power-up, power-down, and entry and exit of any power management state must be controlled to ensure proper operation of the device. Power supplies are arranged into power sequencing groups as shown in Table 19 and Table 20. There are no sequencing requirements between power supplies in the same power sequencing group unless specifically noted. ACPI power states S3 and S4 are not supported.

Table 19. Power Sequencing Group Definitions without Coin Cell Battery

Group	System Power Domain	Voltages
Group A	G3/S5	VDDBT_RTC_G, VDD_18_S5, VDD_33_S5, VDDIO_AUDIO
Group B	S3	VDD_11_S3
Group C	S0	VDDIO, VDDCR_SOC, VDDCR_CPU0, VDDCR_CPU1

Table 20. Power Sequencing Group Definitions with Coin Cell Battery

Group	System Power Domain	Voltages
Group A	G3	VDDBT_RTC_G
Group B	S5	VDD_18_S5, VDD_33_S5, VDDIO_AUDIO
Group C	S3	VDD_11_S3
Group D	S0	VDDIO, VDDCR_SOC, VDDCR_CPU0, VDDCR_CPU1

18.1.1 PWR_GOOD Sequencing Requirements

- All power supplies in all Power Sequencing Groups must be stable and within specification 1 ms before the assertion of PWR_GOOD.
- PWR_GOOD must be deasserted before any supply in any power group is out of specification.

18.1.2 Power-Up Power Sequencing Group Requirements

- All power supplies in Power Sequencing Group A must be stable and within specification before any power supply in Power Sequencing Group B is greater than 10 percent of its specified typical operating value.
- All power supplies in Power Sequencing Group B must be stable and within specification before any power supply in Power Sequencing Group C is greater than 10 percent of its specified typical operating value.
- If using group definitions from Table 20, all power supplies in Power Sequencing Group C must be stable and within specification before any power supply in Power Sequencing Group D is greater than 10 percent of its specified typical operating value.
- All power supplies in Power Sequencing Group C (if using Table 19) or Power Sequencing Group D (if using Table 20) must be stable and within specification 1 ms before the assertion of PWR_GOOD.

18.1.3 Power-Down and Power Failure Power Sequencing Group Requirements

- When SLP_S3_L transitions from 1 to 0, VDDIO, VDDCR_CPU0, VDDCR_CPU1 and VDDCR_SOC must begin to decay.
- When SLP_S5_L transitions from 1 to 0, VDD_11_S3 must begin to decay.
- In an uncontrolled power-down event, when S0 power supplies shut down while S5 supplies are valid, RSMRST_L must be toggled before the system powers on back again to an S0 state.
- No sequencing relationships are required between the Power Sequencing Groups during an uncontrolled power-down or power failure.

18.1.4 Power-Up Power Supply and Signal Sequencing

1. PWROK and RESET_L are open-drain signals. PWROK is connected between the sockets in a 2-socket system as shown below. PWROK is also routed to the SVI3 controller. Similarly, RESET_L is connected together across the die in a socket and between the sockets in a 2-socket system.
2. After PWROK assertion, Startup-VID values are transmitted on SVI3 interfaces.
3. RESET_L remains asserted a minimum of 1 ms after PWROK assertion.
4. PWROK distribution skew between the sockets should be less than 8 ns.

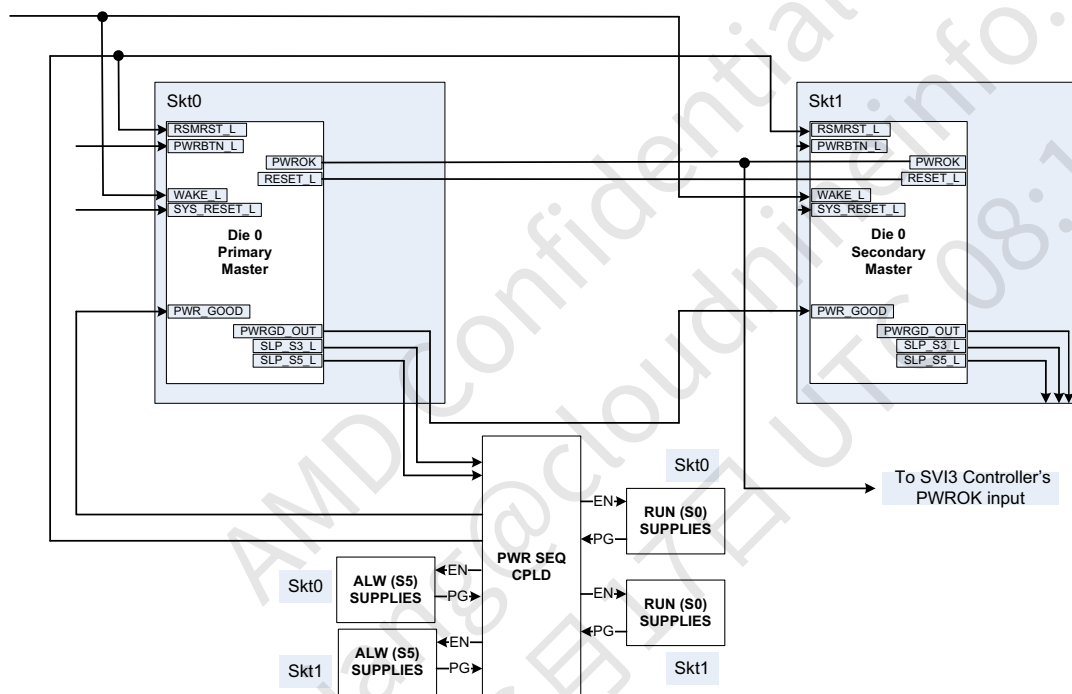


Figure 9. Power and Reset Sequencing Block Diagram

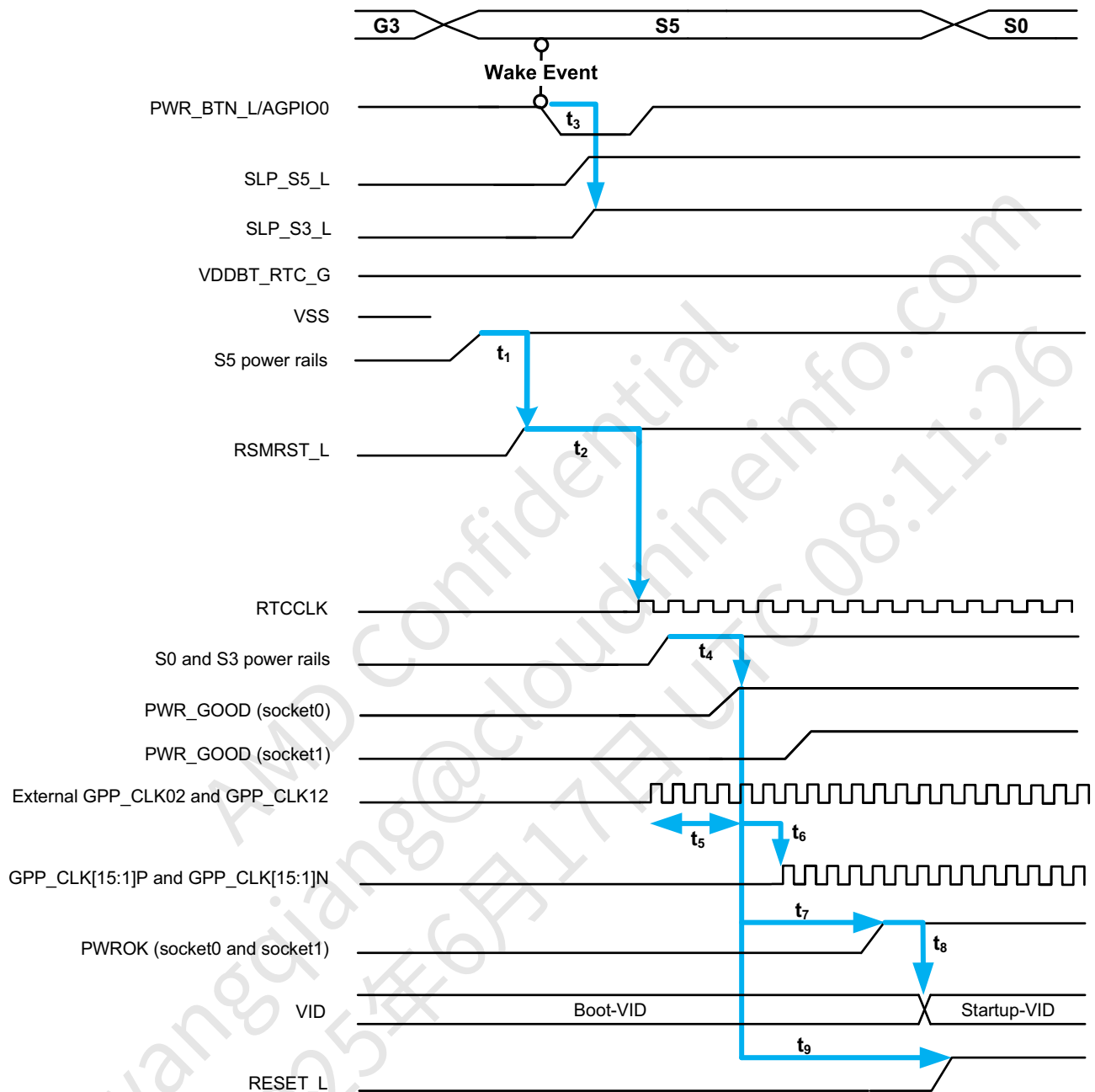


Figure 10. G3->S5->S0 Reset Sequence Timing Diagram

Table 21. G3->S5->S0 Reset Sequencing Timing Parameters

Parameter	Description	Min	Max	Unit
t ₁	From S5 power rails stable to RSMRST_L deassertion.	10	—	ms
t ₂	From RSMRST_L deassertion to RTCCLK assertion.	15	104	
t ₃	From PWR_BTN_L falling to SLP_S3_L deassertion and SLP_S5_L deassertion.	90	360	μs
t ₄	From S0 and S3 power rails stable to PWR_GOOD assertion.	1	—	ms
t ₅	From clock GPP_CLK02 and GPP_CLK12 of both socket0 and socket1 assertion to PWR_GOOD of socket0 assertion.	0	—	ns
t ₆	From PWR_GOOD of socket0 assertion to clock GPP_CLK[15:1]P and GPP_CLK[15:1]N of both socket0 and socket1 assertion.	10.2	26.8	ms
t ₇	1-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	15.0	20.4	
	2-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	26.6	36.7	
t ₈	PWROK to Startup-VID delay (if Startup-VID is sent by processor).	200	—	μs
t ₉	1-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	21.2	28.6	ms
	2-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	32	44	

Notes:

1. RESET_L must remain asserted a minimum of 1 ms after PWROK assertion.
2. SYS_RESET_L must remain high a minimum of 1 ms after rising edge of RSMRST_L.
3. Values in this table assume the hardware reset values of the control registers have not been overwritten.

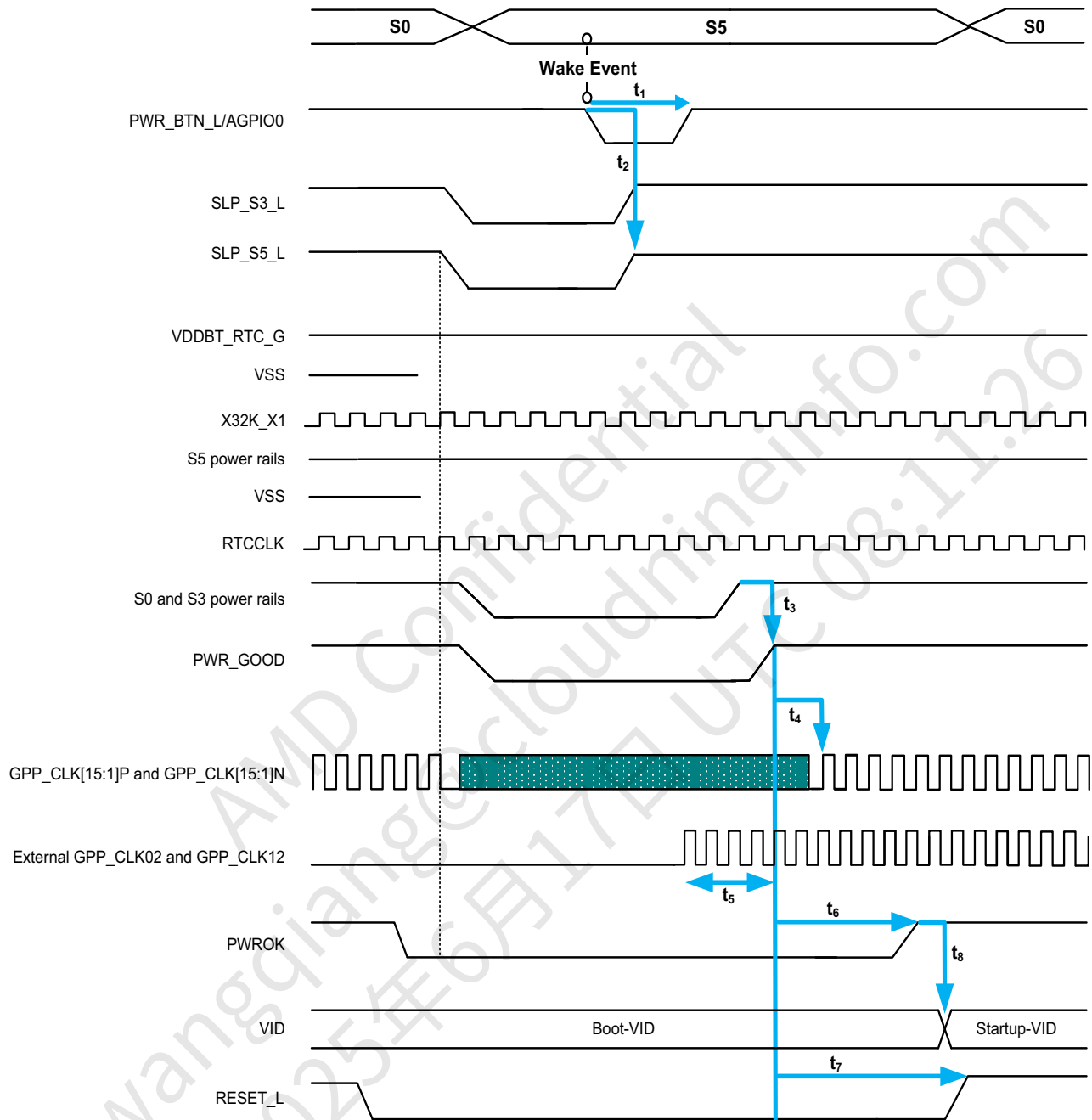


Figure 11. S0->S5->S0 Power Sequencing Timing Diagram

Table 22. S0->S5->S0 Power Sequencing Timing Parameters

Parameter	Description	Min	Max	Unit
t ₁	The minimum assertion time for PWR_BTN_L to wake up the system. Default settings are programmable.	15	—	ms
t ₂	From PWR_BTN_L assertion to SLP_S3_L deassertion and SLP_S5_L deassertion.	14	17	
t ₃	From power rails stable to PWR_GOOD assertion.	1	—	
t ₄	From PWR_GOOD of socket0 assertion to clock GPP_CLK[15:1]P and GPP_CLK[15:1]N of both socket0 and socket1.	10.2	26.8	
t ₅	From clock GPP_CLK02 and GPP_CLK12 of both socket0 and socket1 assertion to PWR_GOOD of socket0 assertion.	0	—	ns
t ₆	1-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	15.0	20.4	ms
	2-socket configuration: From PWR_GOOD of socket0 assertion to PWROK assertion.	26.6	36.7	
t ₇	1-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	21.2	28.6	
	2-socket configuration: From PWR_GOOD of socket0 assertion to RESET_L deassertion.	32	44	
t ₈	PWROK to Startup-VID delay (if Startup-VID is sent by processor).	200	—	μs

Notes:

1. RESET_L remains asserted a minimum of 1 ms after PWROK assertion.
2. SYS_RESET_L must remain high a minimum of 1 ms after rising edge of RSMRST_L.
3. Values in this table assume the hardware reset values of the control registers have not been overwritten.